How to Write AI Agents: a Systematic Literature Review

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Abstract. Agentic AI represents a paradigm shift from reactive systems to autonomous, adaptive, and goal-oriented agents. This shift challenges traditional requirements engineering practices and introduces new forms of specification, particularly following prompt-based and declarative approaches. This paper presents a PRISMA-based systematic literature review focused on how to write or specify AI agents, and that involves a 14-studies dataset. A multi-dimensional analysis is defined considering dimensions such as input & context, output & quality, goal & intent, behavior & control, knowledge & data, interaction & integration, governance & risk. A set of AI agent patterns, guidelines and key recommendations are identified and collected from the literature.

Keywords: AI agents, Agent Patterns, Requirements-Oriented Prompt Engineering, Best Practices, Literature Review.

1 Introduction

Agentic AI is a recent paradigm-shift in artificial intelligence (AI) area, that has shifted from reactive and human-dependent to autonomous and adaptive agents. AI agents are designed to be goal-oriented even in complex and highly changing contexts [1] and are strongly integrated with generative AI technologies, such as large language models (LLMs) and retrieval-augmented generation (RAG). The technical foundations of agentic AI are [1]: Reinforcement Learning (RL); Goal-Oriented Architectures; and Adaptive Control Mechanisms. An AI agent is defined as a system sensible to its environment and able to decide what to do to achieve specified goals [6]. In other words, an AI agent has capabilities such as perception, reasoning (i.e. to think logically to make a judgement), planning, action, memory, learning and adaptation [34]. These agents are supported by automatic learning (AL), natural language processing (NLP) and an orchestration of LLMs in the scope of some agentic AI [6].

On one hand, agentic AI is a framework [6] that supports AI agents which can be specified including requirements [2][36]. A requirement is traditionally defined as a condition or necessary capability for a user or stakeholder to solve a problem or reach a specific objective [18]. Traditional requirements engineering (RE) implies to specify rigorous requirements at the initial phase of the software engineering process, when the exact characteristics of a system are determined, and the concerns and needs of the systems' users are iteratively elaborated [39]. Nevertheless, the innovative

characteristics of agentic AI imply a new approach to RE, i.e., not only producing detailed technical specifications but also guiding AI agents using prompts for defining concisely and unambiguously requirements and instructions for their goals and tasks [5] [26].

For instance, it is possible, without having any software programming background, to interact with agentic AI through a set of prompts in natural language (NL) to define user requirements to create software products [11]. Consequently, software production has being faster using human-AI interaction through prompt engineering discipline [11]. This has stressed the importance of end-user prompt engineering in the broader area of AI engineering [1].

In fact, prompt engineering can be understood as a set of techniques to define properly instructions but also to specify requirements, and, thus, corresponding to a new paradigm of requirements specification named "Requirements-Oriented Prompt Engineering" (ROPE) [27]. In other words, due to distinct applications across various fields (e.g. adaptive software systems, finance, healthcare or robotics) prompts can possibly be defined differently to include distinct technical requirements [1] [26].

Despite the instability and novelty of this research field [19], this paper describes a systematic literature review (SLR) with the purpose of understanding the best techniques and approaches to write AI agents. Furthermore, we propose an analytical framework with dimensions and categories to better analyze and discuss the available literature and a list of key recommendations.

The paper is organized as follows: Section 2 presents the research background; Section 3 the research design of the SLR; Section 4 the plan of the search; Section 5 execute the search; Section 6 the analysis of the results. Finally, Section 7 presents the conclusion and future work do address two key open issues.

2 Background

During software development it is important the elicitation of requirements to gather the objectives of all stakeholders. Functional Requirements (FRs) are detailed descriptions of a system, considering the services provided, how it should react with input data, specific situations and what are the exceptions [52]. On the other hand, quality requirements (QRs) define quality properties of a system, a component, service or function [55] [42]. Among QR we can find: (i) governance [20][53]; (ii) security [53]; (iii) privacy [53]; (iv) scalability [47]; (v) performance [49]; (vi) auditability (and traceability) [40]; (vii) transparency (and explainability) [7]; (viii) trust [53]; (ix) fairness [43]; (x) usability [57]; and (xi) environmental sustainability [3]. Their importance varies across distinct contexts, e.g., healthcare, transportation, software engineering, military and security, manufacturing, smart cities and energy, public administration, tourism and traveling, finance, multi-domain, sales and business, education and scientific research [6].

Because agentic AI systems are complex, the identification of their requirements is difficult [1][6][7]. In fact, agentic AI are systems that coordinate multiple agents, each agent handling a part of a job, and keep them aligned with the goal through an orchestration layer. Popular agentic AI systems are Claude Cowork, CrewAI, LangGraph,

AutoGen, AutoGPT and OpenAI Agents SDK. The core components of an agentic AI system are [6]: memory, reflection and evaluation, perception and world modelling, execution and actuation, planning and reasoning, and orchestration. An AI agent is an entity that senses the environment, decides and takes action to reach a goal, with autonomy [6]. Today autonomy denotes the capability to operate independently without constant external supervision and agency encompasses autonomy plus adaptive behavior, intentionality and goal-directedness, through contextual reasoning, strategic planning, adaptation with actions [31], value alignment and knowledge [41]. In short, the core components of agents are identity, sensors and actuators and are rigorously defined, representing their users [56]. Popular examples of AI agents are virtual assistants, automated workflows and customer service chatbots [47]. Application examples of agentic AI are business process optimization, virtual project managers and supply chain management [47]. This distinction implies distinct requirements for both agentic AI and AI agents.

Actually, AI agents can be written using prompt engineering to define requirements, grounded on several techniques available, e.g. Zero-Shot Prompting, Few-Shot Prompting, Chain-of-Thought (CoT) Prompting, Self-Refine Prompting, Code Prompting [46]. Nevertheless, agentic AI and prompt engineering continue to face several challenges, such as biases, incompleteness, interpretability gaps, ambiguity, factual inaccuracies, lack of safety and ethics [46][50]. In this context, prompts are instructions that operate the stakeholder intents and implementation challenges.

Both agentic AI and AI agents are examples of the modern AI engineering research field. Cao, L. [8] defines AI engineering in three layers: (i) AI engineering techniques (e.g. techniques for AI designs, AI chips, AI algorithms/models, AI systems, and AI computing); (ii) AI system engineering (e.g. AI standardization, AI requirement engineering, AI orchestration, AI system analysis and design, and AI benchmarking); and (iii) AI management and governance (e.g. AI project management, AI risk and compliance, AI privacy and security, and AI ethics). Furthermore, AI engineering uses interdisciplinary contributions, from information systems, software engineering, and intelligent services. In other words, AI engineering develops languages, methodologies, and project management tools for the analysis, design, development and management, evaluation and governance of AI programming. Consequently, because autonomy and agency are core capabilities of actual AI agents, their understanding is essential to identify key elements, defined through prompt engineering in the context of AI engineering.

3 Research Design

The PRISMA methodology [32] was used as a general framework to guide this research. The work by Escallón and Aldea was also used because they presented a methodology with backward and forward paper search, that is valuable in the context of this research [10]. The overall research process is shown in Figure 1. As proposed by Moher et al. [32] a SLR involves three main stages: a phase for planning (Phase-1), a phase for executing that plan (Phase-2), and a phase for analyzing the results from the selected studies (Phase-3).

Phase-2 can include five tasks: (i) extract the studies from databases, (ii) eliminate duplicate studies, (iii) apply inclusion and exclusion criteria, (iv) gather backward and forward citations for considering additional studies, and (v) define the final dataset of selected papers. If, at task-iv, there are new papers found, then the researchers can move backward to task-ii and repeat the process, from task-ii to task-iv, as many times as needed. If any work related to a research task is not present in the currently selected set of references, but it is considered relevant, it should be added to the selected ones. Since Google Scholar is a popular tool and supports the backward and forward analysis of citations, it was the adopted tool.

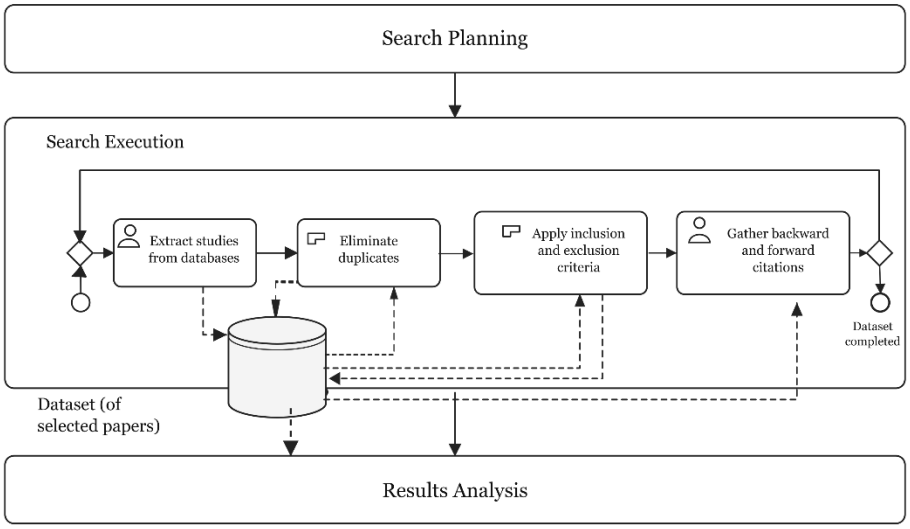


Fig. 1. The overall process of SLR (in BPMN notation).

4 Search Planning (Phase-1)

In the planning phase the design process is followed to perform the SLR according to the selected methodology. Section 4.1 defines the main question that guided this research; Section 4.2 presents the search process; Section 4.3 presents the queries used; and Section 4.4. introduces the analytical framework.

4.1 Research Question

Our study is conducted by one research question (RQ1) which is break-downed with three sub-questions (SQ), specifically:
RQ1: How to write AI agents?
SQ1: What are the relationships between prompt engineering and requirements engineering in the scope of AI agents specification?
SQ2: What are the main prompt patterns to write AI agents?

SQ3: What are the open issues and challenges in future research related to the specification of AI agents?

4.2 Search Process

The following are the databases or scientific repositories to search for relevant papers to answer the questions of this research, because they are well known among the scientific community, and are also used in other SLRs considered as models to follow: ACM Digital Library (dl.acm.org); ScienceDirect/Scopus (www.scopus.com); IEEE Explore (ieeexplore.ieee.org/search/advanced); Elsevier (www.sciencedirect.com); Google Scholar (scholar.google.com); Google Labs (scholar.google.com/scholar_labs/). The inclusion criteria used in this literature review were the following: (i) articles published in the past 5 years; (ii) high-quality studies published in journals or conference proceedings; (iii) studies written in English. The only excluded articles were the ones not relevant for the research field.

4.3. Search Queries

The search query defined to query the papers from these repositories was:

> [[Title: "agent"] OR [Title: "agentic ai"]] AND
> [[Title: guideline] OR [Title: recommendation] OR [Title: style]] AND
> [[Abstract: writ*] OR [Abstract: spec*]] AND
> [E-Publication Date: Past 5 years]

However, we used other queries' variants, considering the convergence between prompt specification and requirements specification, the number of papers found, and the possibility of making a broader search. The search focused on the specification of AI agents and on agentic AI requirements. The time-period of the research was the last five years due to the novelty of the research field.

Some additional queries were used, such as:

> ACM: [[Title: agent] OR [Title: "agentic ai"]] AND [Title: "agent design"] AND
> [E-Publication Date: (01/01/2020 TO 03/31/2026)];
> Elsevier: Title: TITLE-ABS-KEY ("agent design pattern" AND "agentic")
> (Years: 2025);
> IEEE: ("Document Title":agent OR "Document Title":"agentic ai") AND
> ("Document Title":guideline OR "Document Title":recomendation OR
> "Document Title":style) (Date: 01/01/2020 TO 03/15/2026);
> SCOPUS: TITLE-ABS-KEY ("prompt engineering" OR "specification" OR
> "requirements engineering") AND TITLE-ABS-KEY ("agentic") AND
> (LIMIT-TO (EXACTKEYWORD , "Specifications")) (All time);
> GOOGLE LABS: "How can I write a good structure of specifications of an
> AI agent?" and "Show examples of agentic design patterns." (All time)

4.3 Analytical Framework

To better code and analyze the literature and structure the presentation of results we define an analytical framework. This framework captures the multi-layered nature of AI agent specification by combining multiple dimensions, namely: input & context, output & quality, goal & intent, behavior & control, knowledge & data, and interaction & integration, governance & risk. Each of these dimensions, on Table 1, reflects a distinct but interrelated perspective on how AI agents are designed, described, and controlled.

Table 1. Dimensions, Categories and Category Values

Dimension	Category	Category Values (examples)
Input & Context	Input type	User prompt (UP); Event (EV); API request (API); Data stream (DS)
	Structure	Free text (FT); Structured (ST); Multimodal (MM)
	Source	User (USR); System (SYS); External service (EXT); Sensor/IoT (IOT)
	Context scope	Session context (SC); Conversation history (CH); Retrieved context (RAG); Global/system context (GC)
	Quality attributes	Completeness (COMP); Clarity (CLAR); Noise (NOI); Bias (BIAS)
Output & Quality	Output format	Free text (FT); Structured (ST); Reports (REP); Code (CODE)
	Quality attributes	Consistency (CONS); Explainability (EXP); Confidence indicators (CONF)
	Standardization	Schema validation (SV)
Goal & Intent	Goal type	Mission-level (ML); Strategic (STR); Operational (OP); Task-specific (TS); Reactive (REA)
	Abstraction level	High-level (HL); Low-level (LL)
	Granularity	Atomic (AT); Composite (COMP)
	Measurability	KPI-driven (KPI); Qualitative (QUAL)
	Decomposition	Sub-goal decomposition (SGD)
Behavior & Control	Control paradigma	Reactive (REA); Deliberative (DEL); Hybrid (HYB)
	Execution strategy	Planner–Executor (PE); Workflow-driven (WD); LLM-directed (LLM)
	Coordination model	Multi-agent coordination (MAC)
	Adaptation mechanism	Self-reflection loops (SRL); Retry/fallback (RF)
Knowledge & Data	Knowledge source	Internal memory (IM); Knowledge base (KB); Human input (HI)
	Access mechanism	Retrieval (RAG); External APIs/tools (API)
	Data type	Structured (ST); Unstructured (UST)
	Update strategy	Static (STA); Dynamic (DYN); Cached (CAC); Real-time (RT)
Interaction & Integration	Interface type	Chat interface (CHAT); API (API)
	Interaction pattern	Tool invocation (TI); Agent–agent communication (AAC)
	Integration mechanism	Workflow engine (WE); Service orchestration (SO)
	Protocol	Event-driven (ED); Synchronous (SYNC); Asynchronous (ASYNC)
Governance & Risk	Constraint type	Safety (SAFE); Privacy (PRIV); Legal (LEG); Ethical (ETH); Environmental Sustainability (ENV); Sustainable Development (SD)
	Enforcement Mechanism	Guardrails (GR); Human approval (HA)
	Risk type	Risk mitigation (RM) of: hallucinations; security-risk; bias

The **Input & Context** dimension addresses the nature, structure, and acquisition of inputs provided to the agent (user prompts, events, data streams, and contextual signals). The **Output & Quality** dimension considersthe Structure, format, and quality attributes of agent outputs. For example, quality attributes include completeness, clarity, noise and bias. The **Goal & Intent** dimension addresses the nature, abstraction, and operationalization of agent goals, with categories goal type, abstraction level, granularity, measurability and decomposition. The **Behavior & Control** considers the mechanisms governing decision-making, execution, and coordination, for example, the control paradigm, if it is reactive, deliberative or hybrid. The **Knowledge & Data** dimensionconsiders the sources, structure, and access to information used by the agent, mainly including internal sources as internal memory, and external sources as human inputs. The **Interaction & Integration** dimension reflects the modes of communication and integration with users, agents, and systems, such as chat interfaces, APIs, and protocols, e.g. event-driven, synchronous, asynchronous. The **Governance & Risk** dimension includes the constraints and mechanisms ensuring safe, ethical, and compliant behavior, such as risk mitigation approaches, environmental sustainability ethics. These are core dimensions and are described in detail on Table 2.

Other complementary dimensions can be still considered, which can extend constrain, or operationalize the specification depending on the context of use. While not always required in minimal specifications, they are essential in real-world and production environments. They are not analyzed here due to lack of space but we expect to do so in future work. The **Validation & Assurance** dimension focuses on how agent behavior is evaluated and validated, ensuring that specifications can be empirically assessed. The **Lifecycle & Evolution** dimension captures how agent specifications evolve over time, i.e. through update, lifecycle stages, versioning practices, and feedback loops. The **Specification Artifact** dimension characterizes the form in which specifications are expressed, it includes, for example, artifact types such as guidelines, templates, prompts, models.

5 Search Execution (Phase-2)

To execute the search, we followed the tasks and criteria mentioned above and the queries were run in the already mentioned databases, in March 2026. The result was the set of selected papers presented on Table 3. These were the key steps run in this phase: (1st) eliminate duplicates; (2nd) verify if exclusion criteria apply; (3rd) based on the title, select those articles where both inclusion criteria apply, and exclusion criteria do not apply; (4th) repeat the third step but read through the full text; (5th) for each remaining article, review the reference section and repeat steps 2 to 4; and (6th) for relevant references, for each remaining article, Google Scholar is used to review the forward citations and repeat steps 2 to 4. Other literature not included in the SLR is included in this research, due to the novelty of the field, namely two examples of Grey Literature [17][36].

The number of papers screened per database were: 25 for ACM; 1 for Elsevier (+ 1 with Google Scholar); 19 for IEEE Explore (plus 6 with Google Scholar); 11 for Scopus

(+ 2 with Google Scholar); 20 for Google Labs. A total of 46 papers during first search and 9 with Google Scholar. The full list of the selected papers is available at https://github.com/reviewpaperq/papers.

Table 2. Set of selected specification papers.

ID	Title	Authors (abbrev.)	Year
S01	AgentAI: A comprehensive survey on autonomous agents in distributed AI for industry 4.0	Piccialli et al.	2025
S02	AI Agents vs. Agentic AI: A Conceptual taxonomy, applications and challenges	Sapkota et al.	2026
S03	A survey on LLM-based multi-agent systems: workflow, infrastructure, and challenges	Li et al.	2024
S04	Open Agent Specification (Agent Spec): A Unified Representation for AI Agents	Amini et al.	2025
S05	Agent design pattern catalogue: A collection of architectural patterns for foundation model based agents	Liu et al.	2025
S06	A Design-Driven Taxonomy of AI Agentic Patterns	Joyce and Maheshwari	2025
S07	Design of Task Allocation and Decision-Making Styles for AI Agents Based On LLM	Fu and Gou	2025
S08	Towards Designing Conversational Agents for Pair Programming: Accounting for Creativity Strategies and Conversational Styles	Kuttal et al.	2020
S09	Driving Style Alignment for LLM-powered Driver Agent	Yang et al.	2024
S10	PharmaData-Agent: A Specialized Agent for Pharmaceutical Data Analysis	Guan et al.	2025
S11	Talk to your Cobot: faster and more efficient error-handling in a robotic system with a multi-modal Conversational Agent	Schwartz et al.	2024
S12	Trends, Challenges and Processes in Conversational Agent Design: Exploring Practitioners' Views through Semi-Structured Interviews	Sadek et al.	2023
S13	A Requirements Approach for Ecological Health Agentic AI: The Case of Digital Mentor for Occupational Therapists	Hadar et al.	2025
S14	Agentic Design Patterns: A System-Theoretic Framework	Dao et al.	2025

6 Results Analysis (Phase-3)

This section summarizes the results of the papers analyzed, discusses the linkage between prompt engineering and requirements engineering and summarizes key recommendations.

6.1 Multi-dimensional Analysis

The selected papers are structured by the analytical framework introduced above and grounded in the dataset of selected studies. The results are discussed dimension-by-dimension and summarized in Table 3, using abbreviations introduced in Table 1.

Table 3. Coding (Studies x Framework Dimensions)

Study ID	Input & Context	Output & Quality	Goal & Intent	Behavior & Control	Knowledge & Data	Interaction & Integra-tion	Gover-nance & Risk
S01	DS; MM; USR; IOT; GC	ST	STR; OP; COMP; KPI; QUAL	HYB; MAC	KB; DYN	API; AAC; SO	ETH
S02	FT; MM; EXT; GC	FT; ST	ML; TS; AT; COMP; SGD	REA; DEL; MAC; SRL	RAG; API; ST; UST; DYN	API; AAC; TI; SO	SAFE; RM
S03	DS; MM; SYS; GC; COMP	ST; EXP	ML; TS; COMP; SGD	HYB; PE; MAC; SRL	IM; KB; RAG; ST; UST; DYN	AAC; WE; SYNC; ASYNC	RM
S04	UP; FT; ST; USR; EXT; GC	ST	TS; OP; KPI	HYB; PE; WD; MAC	KB; RAG; API; ST; UST	API; SO; SYNC	—
S05	UP; FT; USR; EXT; RAG	FT; EXP; SV	ML; TS; AT; COMP; SGD	DEL; HYB; MAC; SRL	IM; KB; RAG; API; ST; UST; DYN	CHAT; API; AAC; TI; SO; SYNC; ASYNC	SAFE; GR; RM; AT
S06	DS	ST; EXP	REA	REA; DEL; HYB; MAC	IM; DYN	AAC	ETH
S07	UP; USR; RAG	REP; ST	OP; TS; HL; LL; COMP; KPI; QUAL; SGD	HYB; WD	KB; RAG; ST; UST; DYN	SO; SYNC	RM
S08	UP; MM	FT; CODE	TS; COMP	—	HI; UST; DYN	CHAT; SYNC	ETH
S09	DS; FT; ST; HI; SC	FT; ST	OP; TS; REA; AT; COMP; KPI; QUAL	HYB; WD; MAC	IM; HI; ST; UST; DYN	AAC; SO; SYNC	SAFE; RM
S10	UP; FT; ST; MM; USR; SYS; EXT; CH; SC; RAG	REP; CODE; CONS; EXP; SV	TS; OP; COMP; SGD	DEL; HYB; WD; MAC; SRL; RF	KB; RAG; API; ST; UST; DYN	CHAT; TI; SO; SYNC	SAFE; PRIV; GR; RM
S11	UP; EV; DS; MM; USR;	FT	REA; TS; COMP; KPI	REA; WD	KB; RAG; ST; UST; RT	CHAT; TI; SO; SYNC	GR; RM

Study ID	Input & Context	Output & Quality	Goal & Intent	Behavior & Control	Knowledge & Data	Interaction & Integration	Governance & Risk
S12	SYS; SC; RAG UP; USR; GC	REP	STR	WD	HI; UST	AAC; SO	ETH
S13	FT; HI; GC	FT; REP; EXP	STR; OP; TS; COMP; QUAL; SGD	HYB; WD; MAC; SRL	HI; KB; RAG; UST; DYN	CHAT; AAC; SO; ED	ETH; GR; RM
S14	DS; ST; SYS; GC	ST	ML; HL	DEL; PE; MAC	IM; DYN	AAC; SYNC	ETH; RM

Input & Context. The most common data types are user prompt (S01, S03, S06, S09-S11) and data stream (S01-03, S08, S10-S11), with a multimodal structure, and RAG context (S05, S07, S10-11). **Output & Quality**. Most AI agents studied in these papers produce output formats free text (S02, S05, S08, S09, S11, S13) and structured (S01-04, S06, S07, S09, S14). The most valuable quality attribute is explainability (S03, S05, S06, S10, S13), a fact that can be explained by the human difficulties in understanding the complex functioning of LLMs. **Goal & Intent.** Operational (S01, S04, S06, S08, S10, S12) and task-specific (S01-05, S07, S08, S10, S11, S13) are the most common goal types identified by the literature, and their measurability is manly qualitative (S01, S07, S09, S13). **Behavior & Control.** The most common control paradigm is hybrid (S01, S03, S04, S05-7, S09, S13), with a workflow-drive execution strategy (S04, S07, S09-13), a multi-agent coordination model, mentioned in 10 papers (S01-06, S09-10, S13-14), together with self-reflection loops (S02, S03-S05, S10, S13) as an adaptation mechanism. **Knowledge & Data.** Five studies mention RAG technique (S03, S04, S10-11, S13) as a possible knowledge source for AI agents. Four studies mention human-inputs (S08-09, S12-13) and four studies mention external tools (S02, S04-05, S10). This implies that agentic AI is heavily dependent on external data. **Interaction & Integration**. Nine studies explicitly mention agent-agent messaging (S01-03, S05-06, S09, S12-13). Five studies mention chatbots (S05, S08, S10, S11, S13). The main integration mechanism is service orchestration, mentioned by 10 papers, together with a synchronous protocol (S03-05, S07-11, S14). **Governance & Risk**. Ethical concerns are the most frequently mentioned in literature (S01, S06, S08, S12-14), followed by safety (S02, S05, S09-10), together with guardrails as the main enforcement mechanism (S05, S10-11, S13) and the usage of risk mitigation approaches (S03, S05, S07, S09-10, S11, S13-14), e.g. for hallucinations (S02, S05, S11).The ethical and safety concerns and human-in-the-loop supervision is possibly due to the autonomy and agency capabilities of agentic AI (S01). Any papers mention environmental impacts, but paper S13 mentions the possibility of reduction of computational footprints. S05 mentions computational overhead and S01 explicitly mentions industrial financial sustainability.

6.2 Key Recommendations

Some authors propose guidelines to organize these prompts, assuming distinct requirements and prompt patterns for different contexts [44] [48][51]. Other authors stress the demanding requirements of agentic AI [9][47]. An example is scientific research, where the usage of CoT is suggested [16]. Another example referred by Osmani [40, 41], driven by industrial experience, uses NLP in prompts to specify AI agents, with a prompt pattern that results from the analysis of approximately 2500 alternative specifications. Osmany suggests five main guidelines for high-quality AI agents' specifications: (i) start with a high-level vision and then produce an initial draft prompt; (ii) structure the specifications of a product requirement document (or system requirements document), with a well-organized, simple and comprehensive template; (iii) divide tasks into modular prompts and context, to avoid the decreasing quality of outputs due to long and badly structured prompts; (iv) build self-checks, human-in-the-loop and constraints; and (v) test, iterate, and improve the specifications. Also, a sequential and four-phase approach (i.e. specify, plan, subdivide tasks plus implement, and execute plus verify or test) is also mentioned, where, only after the validation at each phase, is possible to move forward. An agent is written with prompts that include requirements, iteration with tools, being the relative importance of each tool context dependent. In fact, the distinct intents of authors for instance, to elicit agentic AI for distinct tasks such as education in computer networks [28], to create test specifications [30], or to produce indirect speech for conversational agents [35][54], all imply different prompt patterns with specific requirements [25].

The analyzed literature have introduced several recommendations to write AI agents, namely:

R1. Before writing an AI agent, you shall understand and identify its description, objectives, requirements, tasks, tools and expected results [14][38];

R2. An AI agent shall start with a meaningful identification [37], that can be structured through a name, description and intent [24];

R3. The chosen prompting technique shall be adapted to the application domain [51] together with proper safety and legal procedures, e.g. in traffic or defense [13], finance [21], and healthcare [13][21][41];

R4. The prompt pattern shall be rigorous, using direct speech (as this speech style is informative) [35][37], and structured in sections, eliciting objectives, tools and requirements [37][38];

R5. A possible sequence of sections shall be: name, description, objectives, requirements, tasks, questioning enforcement, hallucination detection, expected results, available knowledge, tool identification and environmental sustainability;

R6. The first approach and draft prompt shall be produced with the help of Gen AI tools i.e. through a meta-prompt, and be prepared for AI agents' changing objectives [37];

R7. There shall be approaches to overcome the limits of LLMs;

For example, LLMs' context windows are too small to handle complete websites, leading to shallow reasoning and hallucinations [48]. This shall be handled with the creation of a llms.txt file with LLMs' guiding content, such as brief background information, written in markdown language [17].

R8. An AI agent shall consider a human-in-the-loop approach, e.g., for evaluations and domain knowledge requirements [38];
R9. An AI agent shall include guardrails and exception handling, e.g. in industrial production [38];
R10. An AI agent shall include mechanisms for the detection of hallucinations, that can be classified as intrinsic or extrinsic, through automatic data validation or human feedback [24];
R11. You shall adopt an iteration approach due to the evolving reality [16][37] and the positive impacts of training LLMs, using a cycle of test-then-improvements followed by new tests;
R12. You shall add procedures, such as the test of agents [24][25] and the size reduction of specification prompts, to increase the efficiency of AI agents and reduce their environmental impact.

7 Conclusion

This review answers original research questions, regardless of the novelty of the research field, the new discipline of agent engineering [38], that justifies difficulties related to distinct terminology, e.g. requirements, specs, agent design, prompt, prompt patterns [19]. To overcome these difficulties there were a variety of tests with distinct queries until the ones used in this work were identified, together with a proposed analytical framework. According to available literature, there are possibly two main concerns to address in the specification of agentic AI agents: (i) ex-ante specification is not enough to govern autonomous systems [16]; and (ii) the balance between the inclusion of needed requirements in structured prompts, with direct speech, and adequate prompt dimension, to create AI agents is essential for their effectiveness and efficiency [25][32][37]. On the other hand, there are two key open issues for future research: First, the void between industry-based practices and research [29][45], with needed work towards real-time optimization and adaptation [34][41] [12][21], trust [25], human oversight [38], the creation of practical tools and resources practitioners can use [45], and transparency, defined as how understandable and clear users perceive the structure of a system [51]; Second, the need for testing with real-world data [14], considering interdisciplinarity [56][47] and metrics, e.g. for error-handling and hallucinations [6][30], and to understand the contribution of AI agents for sustainable development [16] together with their environmental impacts [20][22][33], for example through the measurement of electricity consumption by distinct prompt patterns.

In summary, this paper presents an SLR and an analytical framework that allowed the identification of key recommendations on how to write AI agents. Theese start from the identification of common components of successful prompt patterns identified in the available literature and are context aware, meaning that some components are more worthwhile being developed in certain application contexts, than others.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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